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AI Trip Budget Planner using Machine Learning

1. Objective

The objective of this project is to develop an AI-based system that predicts travel expenses based on user inputs such as destination, number of days, and budget.

2. Introduction

This project uses Machine Learning techniques to estimate travel costs including hotel, travel, and food expenses. It helps users plan trips efficiently within a given budget.

3. Methodology

The project follows these steps:
- Dataset creation (data.csv)
- Data preprocessing using pandas
- Model training using Linear Regression
- Saving the trained model using pickle
- Building a Flask web application
- Integrating ML model with frontend UI

4. Technologies Used

- Python
- Flask
- Pandas
- Scikit-learn
- HTML/CSS

5. Working of the Project

The user inputs destination, number of days, and budget. The trained ML model predicts hotel cost, travel cost, and food cost. The system also calculates remaining budget and suggests travel mode.

6. Result

The system successfully predicts travel expenses and provides useful insights for budget planning.

7. Conclusion

This project demonstrates the application of Machine Learning in real-world scenarios such as travel planning. It enhances decision-making and improves user experience.

8. Future Scope

- Integration with real-time APIs
- Improved ML models
- Chatbot integration